Project Synopsis: Real-Time Indian Sign Language Detection

A Vision-Based Framework Leveraging Mediapipe Hand Keypoints & Deep Neural Networks

Project Authors / Team Members: **Swaroop** (Reg. No: 245805196) | **Krsaan** (Reg. No: 245805140) | **Rishika** (Reg. No: 245805168) | **Sourish** (Reg. No: 245805168)

1. Problem Statement

Indian Sign Language (ISL) is the primary non-verbal communication medium for millions of deaf and hard-of-hearing individuals across India. However, due to a severe national shortage of certified sign language interpreters, a persistent communication barrier prevents hearing-impaired individuals from interacting seamlessly with broader society in everyday settings. Existing human-dependent interpretation is non-scalable, expensive, and unavailable in real-time emergency, educational, or professional environments. There is an urgent need for an automated, accessible, non-intrusive vision system that translates ISL hand gestures into digital text instantaneously using standard camera hardware.

2. Existing Systems & Research Gap

Existing Systems: Prior research in sign language recognition broadly falls into two categories:
1. *Hardware Sensor Gloves (Cyber-Gloves):* Utilize physical sensors (flex sensors, accelerometers) attached to gloves to capture finger bend and hand orientation. While accurate, sensor gloves are uncomfortable, fragile, expensive, and impractical for daily public use.
2. *Raw RGB Image Convolutional Neural Networks (CNNs):* Computer vision approaches pass raw RGB camera images directly through deep 2D/3D CNNs to classify hand shapes.

Identified Research Gap:

- **Computational Latency & High Resource Footprint:** Raw image-based CNNs require millions of parameters, causing high GPU compute demand and thermal throttling, making them unviable for real-time edge execution on standard CPUs or mobile devices.
- **Environmental & Background Sensitivity:** CNN pixel filters suffer from severe performance degradation when exposed to changing illumination, complex background clutter, user skin tone variations, or clothing artifacts.
- **Lack of Spatial Scale & Origin Invariance:** Existing vision systems degrade significantly if the user shifts hand position within the frame or changes distance relative to the camera lens.

3. Proposed Solution & System Overview

Proposed Solution: To overcome the research gap, this project proposes a lightweight, decoupled two-stage framework:
1. **Spatial Landmark Extraction:** Employs Google's *MediaPipe Hands* solution to extract 21 3D skeletal hand keypoints in real time, converting raw pixel data into lightweight spatial joint coordinates.
2. **Geometric Normalization & Deep Neural Classification:** Applies wrist-origin translation and max-absolute scale normalization to construct a 42-dimensional position- and scale-invariant spatial vector. A Deep Feedforward Neural Network (FNN) then classifies the vector across **35 distinct ISL classes** (Digits 1–9 and Alphabets A–Z) with ultra-low latency.

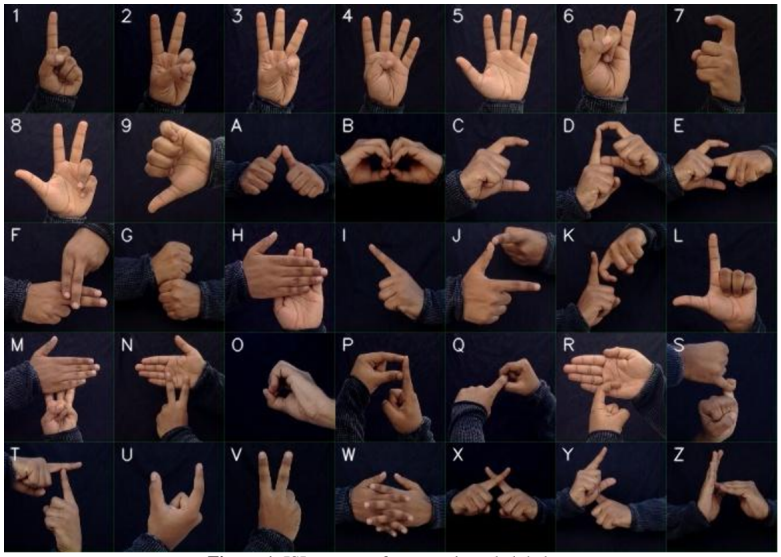


Figure 1. ISL gestures for numeric and alphabets

Figure 1: Comprehensive ISL dataset catalog comprising 35 gesture categories (Digits 1–9 & Alphabets A–Z).

4. System Architecture & Technical Methodology

The end-to-end pipeline processes incoming frames through four sequential modules: Video Acquisition, Keypoint Extraction, Feature Normalization, and Neural Inference.

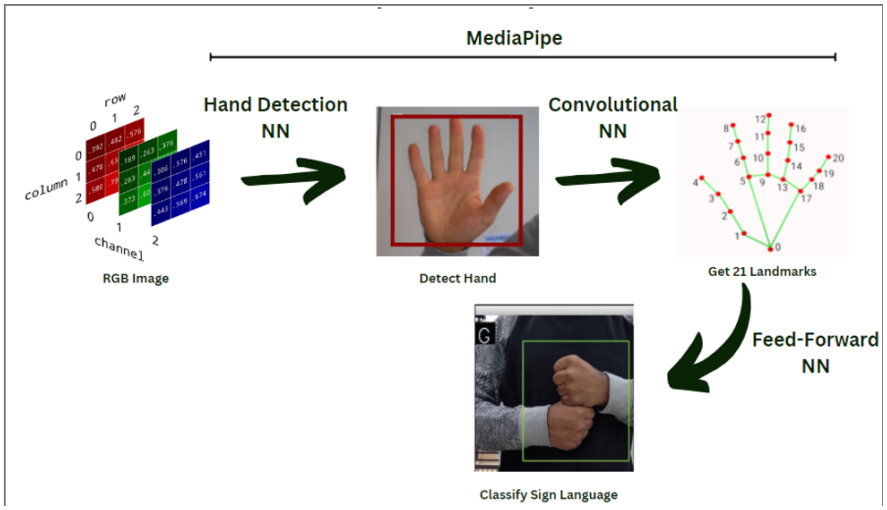


Figure 2: End-to-End System Flowchart detailing frame capture, MediaPipe landmark tracking, keypoint normalization, FNN classification, and screen overlay.

4.1 Hand Landmark Perception & Feature Normalization Pipeline

• **3D Landmark Perception:** MediaPipe Hands detects 21 anatomical joints per hand, yielding 21 2D coordinate pairs *(x_i, y_i)* (42 raw features per hand).
• **Wrist-Origin Translation:** Landmark 0 (wrist) acts as origin anchor *(base_x, base_y)*. All joints are offset: $x'_i = x_i - base_x$, $y'_i = y_i - base_y$.
• **Max-Absolute Normalization:** Coordinates are scaled by maximum absolute magnitude: $x''_i = x'_i / max(|V|)$, $y''_i = y'_i / max(|V|)$, bounding vector values strictly within [-1.0, 1.0].

5. Deep Neural Network Architecture & Performance Evaluation

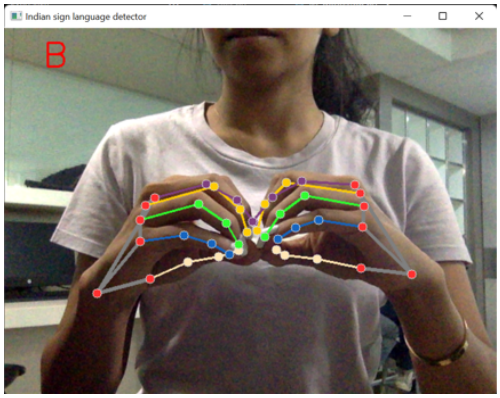
The classification engine is a Deep Feedforward Neural Network (FNN) implemented in Keras / TensorFlow, mapping 42 spatial input features to 35 gesture output classes.

Layer	Layer Type & Config	Units / Dropout Rate	Activation
Input	Dense Input Layer	42 Spatial Features	-
Hidden Layer 1	Dense + Dropout	1,470 units (Dropout: 0.5)	ReLU
Hidden Layer 2	Dense + Dropout	832 units (Dropout: 0.5)	ReLU
Hidden Layer 3	Dense + Dropout	428 units (Dropout: 0.5)	ReLU
Hidden Layer 4	Dense + Dropout	264 units (Dropout: 0.5)	ReLU
Output Layer	Dense Classification	35 Classes	Softmax

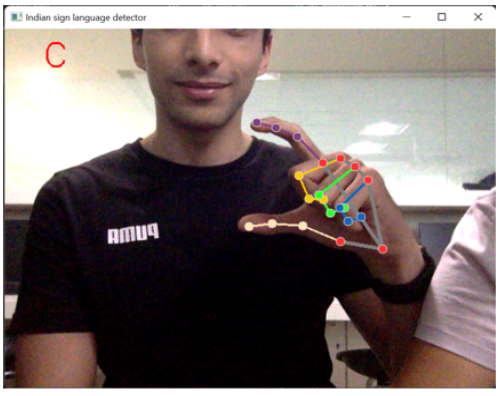
Training Dynamics & Empirical Metrics: Compiled with Sparse Categorical Cross-Entropy loss and Adam adaptive optimizer. 50% Dropout regularization and Early Stopping (patience=2) prevented overfitting, converging at epoch 15. The model achieved **100% Test Accuracy, 1.0 Precision, 1.0 Recall, and 1.0 F1-Score** on the 20% evaluation split.

6. Real-Time Application Deployment & Output

During live webcam execution (isl_detection.py), OpenCV captures video frames, MediaPipe extracts hand landmarks, and the compiled model (model.h5) predicts gesture classes in real time, overlaying recognized labels and hand skeletons on screen.



Real-Time Detection Output 1 (Gesture 'B')



Real-Time Detection Output 2 (Gesture 'C')

7. Summary of Contributions & Future Horizons

Summary of Core Technical Contributions:

• **Ultra-Low CPU Latency:** Compressing image inputs to 42 spatial floats allows millisecond-level inference on CPU edge devices without dedicated GPUs.

• **Environmental Immunity:** Isolating joint geometry from RGB pixels makes recognition immune to lighting changes, skin tone, background noise, or clothing.

• **Position & Scale Invariance:** Wrist-relative offset and max-abs vector scaling preserve high accuracy across camera distances and user hand sizes.

Future Horizons: Integrating Recurrent Neural Networks (RNNs / LSTMs / Transformers) for continuous multi-gesture ISL sentence tracking, Text-to-Speech (TTS) voice synthesis, and cross-platform desktop/mobile GUI packaging.